Unit ENV-III

SKU:U001-C













Description

Unit ENV-III is an environmental sensor that integrates SHT30 and QMP6988 internally to detect temperature, humidity, and atmospheric pressure data. The SHT30 is a high-precision, low-power digital temperature and humidity sensor that supports the I2C interface (SHT30: 0x44, QMP6988: 0x70). The QMP6988 is an absolute pressure sensor designed for mobile applications, offering high accuracy and stability, suitable for projects involving environmental data collection and detection.

Features

- Easy to use
- High accuracy
- I2C communication interface
- HY2.0-4P interface, supports platforms UIFlow and Arduino.
- 2 x LEGO compatible holes

Includes

- 1 x Unit ENV-III
- 1 x HY2.0-4P Grove connection cable (20cm)

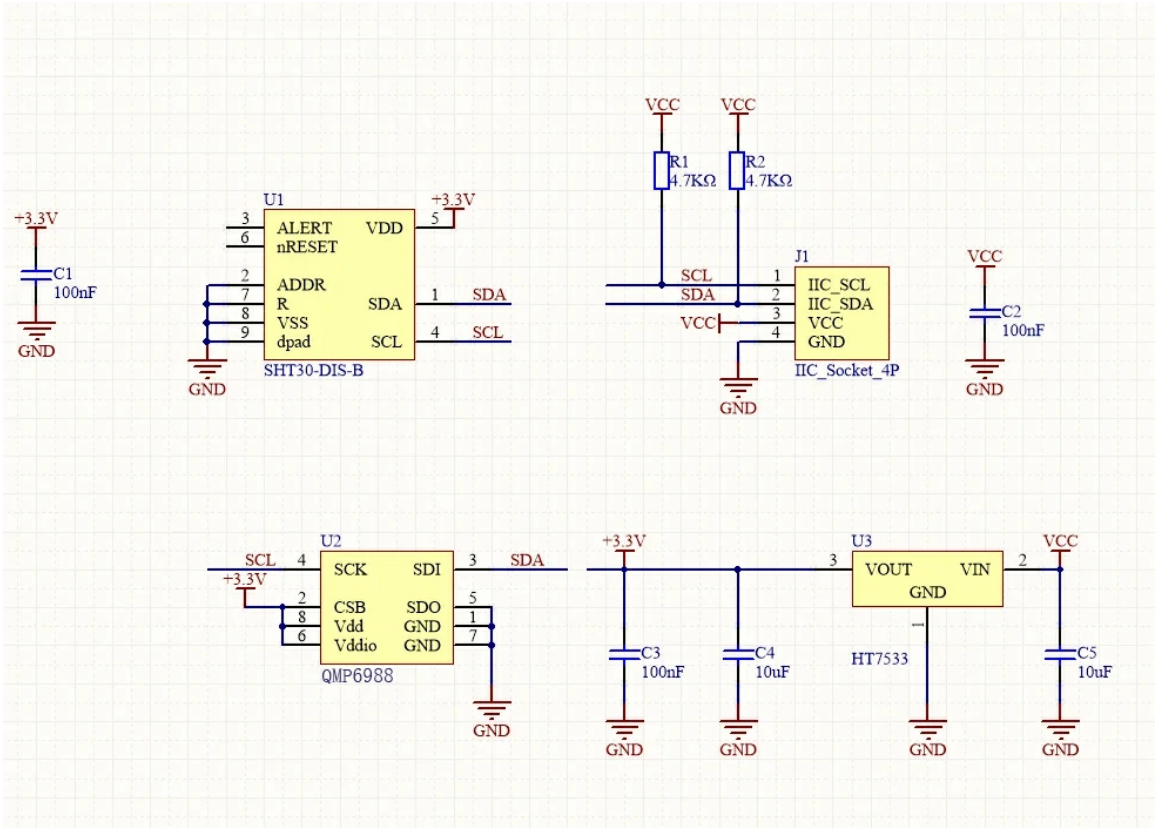
Applications

- Weather stations
- Grain storage environment monitoring

Specifications

Specification	Parameter
Maximum Temperature Range	-40 ~ 120 ℃
Highest Measurement Accuracy	0 ~ 60 ℃/±0.2℃
Humidity Range/Error	10 ~ 90 %RH / ±2%
Maximum Pressure Value/Resolution/Error	300 ~ 1100hPa / 0.06Pa / ±3.9Pa
Communication Protocol	I2C: SHT30 (0x44), QMP6988 (0x70)
Operating Temperature	0 ~ 60°C
Case Material	Plastic (PC)
Product Size	32.0 x 24.0 x 8.0mm
Product Weight	6.2g
Package Size	138.0 x 93.0 x 9.0mm
Gross Weight	14.8g

Schematics

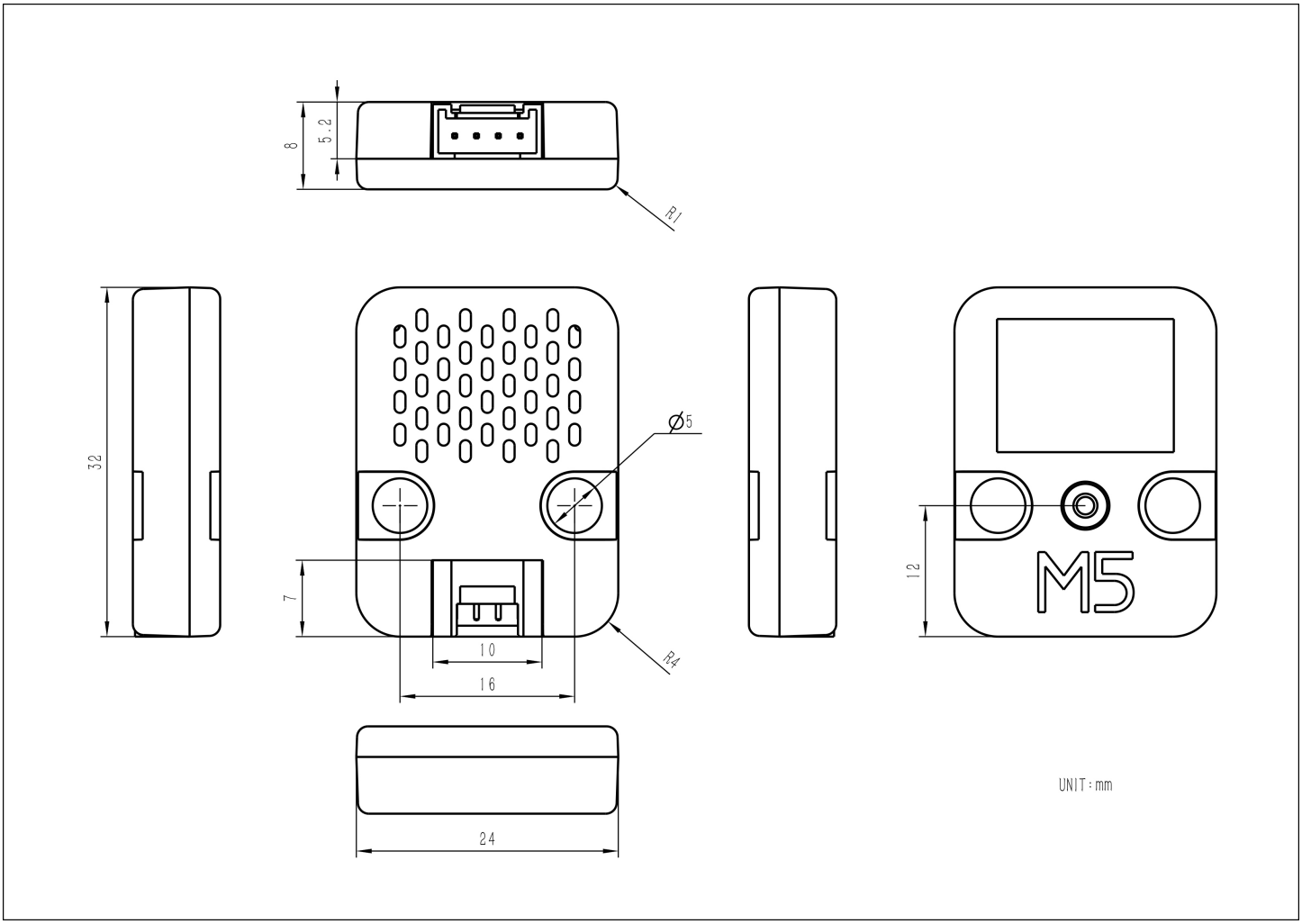


PinMap

Unit ENV-III

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	SDA	SCL

Model Size



Structure

- Unit ENV-III Structure Files

Datasheets

- QMP6988
- SHT30

Softwares

Arduino

- Unit ENV-III Arduino Tutorial
- Unit ENV-III Test Example with M5Atom
- Unit ENV-III Test Example with M5AtomS3
- Unit ENV-III Test Example with M5AtomS3Lite
- Unit ENV-III Test Example with M5Core
- Unit ENV-III Test Example with M5Core2
- Unit ENV-III Test Example with M5StickC
- Unit ENV-III Test Example with M5StickCPlus

UiFlow1

- Unit ENV-III UiFlow1 Docs

UiFlow2

- Unit ENV-III UiFlow2 Docs

Product Comparison

Product Compare					
	Unit ENV	**Unit ENV-II**	**Unit ENV-III**	**Unit ENV-IV**	**Unit ENV-Pro**
Sensor	DHT12+BMP280	SHT30+BMP280	SHT30+QMP6988	SHT40+BMP280	BME688
Data Type Detected	Temperature+Pressure+Humidity	Temperature+Pressure+Humidity	Temperature+Pressure+Humidity	Temperature+Pressure+Humidity	Temperature+Pressure+Humidity+Gas
Interface	I2C	I2C	I2C	I2C	I2C
I2C Address	DHT12:0x5C BMP280:0x76	SHT30:0x44 BMP280:0x76	SHT30:0x44 QMP6988:0x70	SHT40:0x44 BMP280:0x76	BME688:0x77
Accuracy	Temp: ±0.5℃ Humidity: ±5%RH Pressure: ±0.12 hPa	Temp: ±0.2℃ Humidity: ±2%RH Pressure: ±0.12 hPa	Temp: ±0.2℃ Humidity: ±2%RH Pressure: ±0.12 hPa	Temp: ±0.2℃ Humidity: ±1.8%RH Pressure: ±0.12 hPa	Temp: ±0.5℃ Humidity: ±3%RH Pressure: ±0.12 hPa
Range	Temp: -20~60℃ Humidity: 20~95%RH Pressure: 300~1100 hPa	Temp: -40~125℃ Humidity: 0~100%RH Pressure: 300~1100 hPa	Temp: -40~125℃ Humidity: 0~100%RH Pressure: 300~1100 hPa	Temp: -40~125℃ Humidity: 0~100%RH Pressure: 300~1100 hPa	Scan speed: 10.8 seconds